Taylor and Maclaurin Series

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: The Taylor series formula

State the general Taylor series formula for $f(x)$ centered at $x = a$.

$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots$.

2 [Foundational] • Target: The Taylor series formula

Find the Taylor series for $f(x) = \ln x$ centered at $a = 1$, up through the $(x-1)^3$ term.

$f(1) = 0$; $f'(x) = \frac{1}{x}$, $f'(1) = 1$; $f''(x) = -\frac{1}{x^2}$, $f''(1) = -1$; $f'''(x) = \frac{2}{x^3}$, $f'''(1) = 2$. Series: $0 + (x-1) - \frac{1}{2}(x-1)^2 + \frac{2}{6}(x-1)^3 = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3}$.

3 [Foundational] • Target: Common Maclaurin series

State the Maclaurin series for e^x, $\sin x$, and $\cos x$.

$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$. $\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$.
$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$.

4 [Foundational] • Target: Common Maclaurin series

Use the Maclaurin series for e^x to find the Maclaurin series for e^{-x^2}.

Substitute $x \to -x^2$: $e^{-x^2} = \sum_{n=0}^{\infty} \frac{(-x^2)^n}{n!} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{n!} = 1 - x^2 + \frac{x^4}{2} - \cdots$.

5 [Foundational] • Target: Using known series to build new series

Use the Maclaurin series for $\sin x$ to find the first three nonzero terms of the Maclaurin series for $x\sin x$.

$x\sin x = x\left(x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots\right) = x^2 - \frac{x^4}{6} + \frac{x^6}{120} - \cdots$.

6 [Foundational] • Target: Using known series to build new series

Use the known Taylor series for $\cos x$ to estimate $\cos(0.1)$ using the first three nonzero terms.

$\cos(0.1) \approx 1 - \frac{(0.1)^2}{2} + \frac{(0.1)^4}{24} = 1 - 0.005 + 0.0000042 \approx 0.995004$.

7 [Foundational] • Target: The Lagrange error bound

State the Lagrange Error Bound formula for the remainder $R_n(x)$ of a Taylor polynomial of degree n.

$|R_n(x)| \le \frac{M}{(n+1)!}|x-a|^{n+1}$, where M is an upper bound for $|f^{(n+1)}(z)|$ on the interval between a and x.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: The Lagrange error bound**

Use the Lagrange Error Bound to estimate the maximum error when approximating $\sin(0.5)$ with the Taylor polynomial $P_3(x) = x - \frac{x^3}{6}$ (degree 3, centered at 0), given $|f^{(4)}(z)| \le 1$ for all z.

$|R_3(0.5)| \le \frac{1}{4!}|0.5|^4 = \frac{0.0625}{24} \approx 0.0026$.

9 **[Intermediate]** • **Target: Radius of convergence of Taylor series**

State the radius of convergence of the Maclaurin series for e^x, $\sin x$, and $\cos x$.

All three series have radius of convergence $R = \infty$ — they converge for all real x.

10 **[Intermediate]** • **Target: Visualizing Taylor polynomial approximations**

Compare $f(x) = \sin x$ to its degree-5 Maclaurin polynomial $P_5(x) = x - \frac{x^3}{6} + \frac{x^5}{120}$ on $[-\pi, \pi]$, and describe where the approximation is best and worst.

The approximation is excellent near $x = 0$ (where the curves are nearly indistinguishable) and progressively worsens toward $x = \pm\pi$, since Taylor polynomials are most accurate near the center of expansion and lose accuracy farther away.

11 **[Intermediate]** • **Target: Visualizing Taylor polynomial approximations**

Compare $f(x) = e^x$ to its degree-2 Maclaurin polynomial $P_2(x) = 1 + x + \frac{x^2}{2}$ on $[-2, 2]$, and identify visually where P_2 underestimates or overestimates e^x.

Since e^x is convex (concave up) and its Taylor polynomial is tangent to it at $x = 0$ up to second order, $P_2(x)$ tends to underestimate e^x away from the center, with the gap growing faster on the positive side due to the exponential's rapid growth.

12 **[Intermediate]** • **Target: Building Taylor series at a nonzero center**

Find the Taylor series for $f(x) = \cos x$ centered at $a = \frac{\pi}{2}$, up through the cubic term.

$f(\frac{\pi}{2}) = 0$; $f'(x) = -\sin x$, $f'(\frac{\pi}{2}) = -1$; $f''(x) = -\cos x$, $f''(\frac{\pi}{2}) = 0$; $f'''(x) = \sin x$, $f'''(\frac{\pi}{2}) = 1$. Series: $-(x - \frac{\pi}{2}) + \frac{1}{6}(x - \frac{\pi}{2})^3 - \cdots$.

13 **[Intermediate]** • **Target: Applying the Lagrange Error Bound to bound accuracy requirements**

Using the degree-4 Maclaurin polynomial for $\cos x$ and the bound $|f^{(5)}(z)| \le 1$, find the maximum error in approximating $\cos(0.3)$.

$|R_4(0.3)| \le \frac{1}{5!}|0.3|^5 = \frac{0.00243}{120} \approx 0.0000203$ — an extremely small error, confirming the Maclaurin polynomial is highly accurate this close to the center.

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14 **[Intermediate]** • **Target: The binomial series**

State the binomial series for $(1+x)^k$ up through the quadratic term, and use it to find the Maclaurin series for $\sqrt{1+x}$ (i.e. $k=\frac{1}{2}$) up through the quadratic term.

$(1+x)^k = 1 + kx + \frac{k(k-1)}{2}x^2 + \cdots$. With $k=\frac{1}{2}$: $\sqrt{1+x} = 1 + \frac{1}{2}x + \frac{(1/2)(-1/2)}{2}x^2 + \cdots = 1 + \frac{x}{2} - \frac{x^2}{8} + \cdots$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: The binomial series** • **Cross-Lesson: 3.02 Vector-Valued Functions**

Use the series found above to approximate $\sqrt{1.1}$ using the first three terms.

$\sqrt{1.1} \approx 1 + \frac{0.1}{2} - \frac{(0.1)^2}{8} = 1 + 0.05 - 0.00125 = 1.04875$ (true value ≈ 1.04881).

16 **[Challenging]** • **Target: Deriving new Maclaurin series from known ones** • **Cross-Lesson: 4.01 Polar Curves and Derivatives**

Use the known series for e^x to find the Maclaurin series for $\cosh x = \frac{e^x + e^{-x}}{2}$, up through the quartic term.

$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \cdots$; $e^{-x} = 1 - x + \frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{24} - \cdots$. Averaging cancels odd terms: $\cosh x = 1 + \frac{x^2}{2} + \frac{x^4}{24} + \cdots$.

17 **[Challenging]** • **Target: Using known series to build new series** • **Cross-Lesson: 4.02 Area in Polar Coordinates**

For $x = t^3,\ y = t^2$, find $\frac{dy}{dx}$ at $t = 1$.

Method · Parametric Tangent Slope Evaluation: Step 1 · Differentiate components with respect to t: $\frac{dx}{dt} = 3t^2,\quad \frac{dy}{dt} = 2t$. Step 2 · Form derivative ratio: $\frac{dy}{dx} = \frac{2t}{3t^2} = \frac{2}{3t}\quad (te0)$. Step 3 · Evaluate at $t = 1$: $\left.\frac{dy}{dx}\right|_{t=1} = \frac{2}{3(1)} = \mathbf{\frac{2}{3}}$.

18 **[Challenging]** • **Target: The Lagrange error bound** • **Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests**

For $x = t^2,\ y = 2t$, find $\frac{dy}{dx}$.

Theorem (Parametric Differentiation): Step 1 · Differentiate components with respect to t: $\frac{dx}{dt} = \frac{d}{dt}[t^2] = 2t,\quad \frac{dy}{dt} = \frac{d}{dt}[2t] = 2$. Step 2 · Compute the derivative ratio: $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2}{2t} = \mathbf{\frac{1}{t}}\quad (te0)$.

19 **[Challenging]** • **Target: Radius of convergence of Taylor series** • **Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests**

Using $\sin(3x) = \sum_{n=0}^{\infty} \frac{(-1)^n (3x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^5.

Final Answer: $\frac{81}{40}$. With $2n+1 = 5$, $n = 2$: coefficient $= \frac{(-1)^2 3^5}{5!} = (1)\cdot\frac{243}{120} = \frac{81}{40}$.

20 **[Challenging]** • **Target: Visualizing Taylor polynomial approximations** • **Cross-Lesson: 6.01 Power Series and Interval of Convergence**

Using $\sin(3x) = \sum_{n=0}^{\infty} \frac{(-1)^n (3x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3.

Final Answer: $-\frac{9}{2}$. With $2n+1 = 3$, $n = 1$: coefficient $= \frac{(-1)^1 3^3}{3!} = (-1)\cdot\frac{27}{6} = -\frac{9}{2}$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] Target: The Taylor series formula

Using $e^{3x} = \sum_{n=0}^{\infty} \frac{(3x)^n}{n!}$, find the coefficient of x^5.

a. $\frac{121}{40}$

b. $\frac{81}{8}$

c. $\frac{1}{40}$

d. $\frac{81}{40}$ ✓

The coefficient of x^n is $\frac{3^n}{n!}$, so the coefficient of x^5 is $\frac{3^5}{5!} = \frac{243}{120} = \frac{81}{40}$.

22 [Foundational] Target: Common Maclaurin series

Using $e^{6x} = \sum_{n=0}^{\infty} \frac{(6x)^n}{n!}$, find the coefficient of x^2.

a. 18 ✓

b. 19

c. 3

d. 36

The coefficient of x^n is $\frac{6^n}{n!}$, so the coefficient of x^2 is $\frac{6^2}{2!} = \frac{36}{2} = 18$.

23 [Foundational] Target: Using known series to build new series

Using $e^{8x} = \sum_{n=0}^{\infty} \frac{(8x)^n}{n!}$, find the coefficient of x^4.

a. $\frac{512}{3}$ ✓

b. $\frac{2048}{3}$

c. $\frac{515}{3}$

d. $\frac{1}{3}$

The coefficient of x^n is $\frac{8^n}{n!}$, so the coefficient of x^4 is $\frac{8^4}{4!} = \frac{4096}{24} = \frac{512}{3}$.

24 [Foundational] Target: The Lagrange error bound

Using $e^{8x} = \sum_{n=0}^{\infty} \frac{(8x)^n}{n!}$, find the coefficient of x^3.

a. $\frac{4}{3}$

b. 256

c. $\frac{259}{3}$

d. $\frac{256}{3}$ ✓

The coefficient of x^n is $\frac{8^n}{n!}$, so the coefficient of x^3 is $\frac{8^3}{3!} = \frac{512}{6} = \frac{256}{3}$.

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25 [Foundational] • Target: Radius of convergence of Taylor series

Using $e^{2x} = \sum_{n=0}^{\infty} \frac{(2x)^n}{n!}$, find the coefficient of x^2.

a. 4

b. 3

c. 2 ✓

d. 1

The coefficient of x^n is $\frac{2^n}{n!}$, so the coefficient of x^2 is $\frac{2^2}{2!} = \frac{4}{2} = 2$.

26 [Foundational] • Target: Visualizing Taylor polynomial approximations

Using $e^{6x} = \sum_{n=0}^{\infty} \frac{(6x)^n}{n!}$, find the coefficient of x^5.

a. 324

b. $\frac{324}{5}$ ✓

c. $\frac{329}{5}$

d. $\frac{1}{20}$

The coefficient of x^n is $\frac{6^n}{n!}$, so the coefficient of x^5 is $\frac{6^5}{5!} = \frac{7776}{120} = \frac{324}{5}$.

27 [Foundational] • Target: Building Taylor series at a nonzero center

Using $e^{8x} = \sum_{n=0}^{\infty} \frac{(8x)^n}{n!}$, find the coefficient of x^2.

a. 32 ✓

b. 64

c. 33

d. 4

The coefficient of x^n is $\frac{8^n}{n!}$, so the coefficient of x^2 is $\frac{8^2}{2!} = \frac{64}{2} = 32$.

28 [Foundational] • Target: Applying the Lagrange Error Bound to bound accuracy requirements

Using $\sin(6x) = \sum_{n=0}^{\infty} \frac{(-1)^n (6x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3.

a. -35

b. 36

c. -36 ✓

d. 108

With $2n+1 = 3$, $n = 1$: coefficient $= \frac{(-1)^1 6^3}{3!} = (-1) \cdot \frac{216}{6} = -36$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: The binomial series

Using $\sin(5x) = \sum_{n=0}^{\infty} \frac{(-1)^n (5x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3.

a. $\frac{125}{2}$

b. $-\frac{125}{6}$ ✓

c. $\frac{125}{6}$

d. $-\frac{119}{6}$

With $2n+1=3$, $n=1$: coefficient $= \frac{(-1)^1 5^3}{3!} = (-1)\cdot\frac{125}{6} = -\frac{125}{6}$.

30 [Intermediate] • Target: Deriving new Maclaurin series from known ones

Using $\sin(4x) = \sum_{n=0}^{\infty} \frac{(-1)^n (4x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3.

a. $-\frac{29}{3}$

b. $\frac{32}{3}$

c. 32

d. $-\frac{32}{3}$ ✓

With $2n+1=3$, $n=1$: coefficient $= \frac{(-1)^1 4^3}{3!} = (-1)\cdot\frac{64}{6} = -\frac{32}{3}$.

31 [Intermediate] • Target: Common Maclaurin Series ($e^x, \sin x, \cos x$)

Using $\sin(2x) = \sum_{n=0}^{\infty} \frac{(-1)^n (2x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3.

a. 4

b. $-\frac{1}{3}$

c. $\frac{4}{3}$

d. $-\frac{4}{3}$ ✓

With $2n+1=3$, $n=1$: coefficient $= \frac{(-1)^1 2^3}{3!} = (-1)\cdot\frac{8}{6} = -\frac{4}{3}$.

32 [Intermediate] • Target: Constructing Taylor Polynomials from Derivatives

Using $\sin(9x) = \sum_{n=0}^{\infty} \frac{(-1)^n (9x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^3.

a. $-\frac{241}{2}$

b. $-\frac{243}{2}$ ✓

c. $\frac{729}{2}$

d. $\frac{243}{2}$

With $2n+1=3$, $n=1$: coefficient $= \frac{(-1)^1 9^3}{3!} = (-1)\cdot\frac{729}{6} = -\frac{243}{2}$.

33 [Intermediate] • Target: Manipulating Known Taylor Series

Using $\sin(5x) = \sum_{n=0}^{\infty} \frac{(-1)^n (5x)^{2n+1}}{(2n+1)!}$, find the coefficient of x^5.

a. $-\frac{625}{24}$

b. $\frac{649}{24}$

c. $\frac{625}{24}$ ✓

d. $\frac{3125}{24}$

With $2n+1=5$, $n=2$: coefficient $= \frac{(-1)^2 5^5}{5!} = (1)\cdot\frac{3125}{120} = \frac{625}{24}$.

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34 **[Intermediate]** • **Target: Lagrange Error Bound (Taylor's Remainder Theorem)**

Use the Maclaurin series for e^u (with $u = 3x^2$) to find the coefficient of x^8 in the Maclaurin series for e^{3x^2}.

a. $\frac{2187}{8}$

b. $\frac{9}{4480}$

c. $\frac{35}{8}$

d. $\frac{27}{8}$ ✓

Substituting $u = 3x^2$ into $e^u = \sum \frac{u^n}{n!}$: the x^8 term comes from $n = 4$: $\frac{(3x^2)^4}{4!} = \frac{81}{24}x^8$. Coefficient: $\frac{27}{8}$.

35 **[Intermediate]** • **Target: The Taylor series formula**

Use the Maclaurin series for e^u (with $u = 2x^2$) to find the coefficient of x^8 in the Maclaurin series for e^{2x^2}.

a. $\frac{2}{3}$ ✓

b. $\frac{1}{2520}$

c. $\frac{5}{3}$

d. $\frac{32}{3}$

Substituting $u = 2x^2$ into $e^u = \sum \frac{u^n}{n!}$: the x^8 term comes from $n = 4$: $\frac{(2x^2)^4}{4!} = \frac{16}{24}x^8$. Coefficient: $\frac{2}{3}$.

36 **[Intermediate]** • **Target: Common Maclaurin series**

Using the Lagrange Error Bound $|R_n(x)| \le \frac{M}{(n+1)!}|x-a|^{n+1}$ with $M = 6$, $n = 2$, and $|x-a| = 1$, find the error bound.

a. 2

b. $\frac{7}{6}$

c. 3

d. 1 ✓

$|R_2(x)| \le \frac{6}{(2+1)!}(1)^{2+1} = \frac{6}{6} = 1$.

37 **[Intermediate]** • **Target: Using known series to build new series**

Using the Lagrange Error Bound $|R_n(x)| \le \frac{M}{(n+1)!}|x-a|^{n+1}$ with $M = 5$, $n = 3$, and $|x-a| = 1$, find the error bound.

a. $\frac{5}{6}$

b. $\frac{5}{24}$ ✓

c. $\frac{29}{24}$

d. $\frac{1}{4}$

$|R_3(x)| \le \frac{5}{(3+1)!}(1)^{3+1} = \frac{5}{24} = \frac{5}{24}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: The Lagrange error bound • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Using the Lagrange Error Bound $|R_n(x)| \le \frac{M}{(n+1)!}|x-a|^{n+1}$ with $M = 4$, $n = 2$, and $|x-a| = 1$, find the error bound.

a. $\frac{5}{3}$

b. 2

c. $\frac{5}{6}$

d. $\frac{2}{3}$ ✓

$|R_2(x)| \le \frac{4}{(2+1)!}(1)^{2+1} = \frac{4}{6} = \frac{2}{3}$.

39 [Challenging] • Target: Radius of convergence of Taylor series • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Using $\cosh(8x) = \frac{e^{8x}+e^{-8x}}{2} = \sum_{n=0}^{\infty} \frac{(8x)^{2n}}{(2n)!}$, find the coefficient of x^6.

a. $\frac{32768}{15}$

b. $\frac{16384}{315}$

c. $\frac{16384}{45}$ ✓

d. $\frac{16429}{45}$

With $2n = 6$: coefficient $= \frac{8^6}{6!} = \frac{262144}{720} = \frac{16384}{45}$.

40 [Challenging] • Target: Visualizing Taylor polynomial approximations • Cross-Lesson: 6.01 Power Series and Interval of Convergence

Using $\cosh(4x) = \frac{e^{4x}+e^{-4x}}{2} = \sum_{n=0}^{\infty} \frac{(4x)^{2n}}{(2n)!}$, find the coefficient of x^4.

a. $\frac{35}{3}$

b. $\frac{128}{3}$

c. $\frac{32}{3}$ ✓

d. $\frac{32}{15}$

With $2n = 4$: coefficient $= \frac{4^4}{4!} = \frac{256}{24} = \frac{32}{3}$.

41 [Challenging] • Target: Common Maclaurin Series ($e^x, \sin x, \cos x$) • Cross-Lesson: 1.01 Integration by Parts

What is the coefficient of x^4 in the Maclaurin series for e^{-2x}?

a. $\frac{2}{3}$ ✓

b. $\frac{16}{24} = \frac{2}{3}$

c. $-\frac{2}{3}$

d. $\frac{16}{4}$

General term for e^u is $\frac{u^n}{n!}$. For $u = -2x$ and $n = 4$: $\frac{(-2x)^4}{4!} = \frac{16x^4}{24} = \frac{2}{3}x^4$.

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42 [Challenging] • Target: Constructing Taylor Polynomials from Derivatives • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

Which formula gives the n-th degree Taylor polynomial $T_n(x)$ centered at $x = c$?

a. $\sum_{k=0}^{n} \frac{f^{(k)}(c)}{k!}(x-c)^k$ ✓

b. $\sum_{k=0}^{n} f^{(k)}(c)(x-c)^k$

c. $\sum_{k=0}^{n} \frac{f(c)}{k!}(x-c)^k$

d. $\frac{f^{(n)}(c)}{n!}(x-c)^n$

By definition, the n-th degree Taylor polynomial is $T_n(x) = f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \ldots + \frac{f^{(n)}(c)}{n!}(x-c)^n$.

43 [Challenging] • Target: Constructing Taylor Polynomials from Derivatives • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

If a function f has $f(2) = 5, f'(2) = -3, f''(2) = 8$, what is $T_2(2.1)$?

a. 4.74 ✓

b. 4.70

c. 4.82

d. 5.04

$T_2(x) = 5 - 3(x-2) + \frac{8}{2}(x-2)^2 = 5 - 3(x-2) + 4(x-2)^2$. At $x = 2.1$: $5 - 3(0.1) + 4(0.01) = 5 - 0.3 + 0.04 = 4.74$.

44 [Challenging] • Target: Manipulating Known Taylor Series • Cross-Lesson: 3.01 Parametric Equations

Find the Maclaurin series for $\frac{\sin x - x}{x^3}$.

a. $-\frac{1}{6} + \frac{x^2}{120} - \ldots$ ✓

b. $\frac{1}{6} - \frac{x^2}{120} + \ldots$

c. $1 - \frac{x^2}{6} + \ldots$

d. $-\frac{x^3}{6} + \ldots$

$\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \ldots \Rightarrow \sin x - x = -\frac{x^3}{6} + \frac{x^5}{120} - \ldots$ Dividing by x^3 yields $-\frac{1}{6} + \frac{x^2}{120} - \ldots$

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45 [Challenging] • Target: Lagrange Error Bound (Taylor's Remainder Theorem) • Cross-Lesson: 3.02 Vector-Valued Functions

What is the Lagrange Error Bound formula for $|R_n(x)| = |f(x) - P_n(x)|$ centered at c?

a. $|R_n(x)| \leq \frac{M}{(n+1)!}|x-c|^{n+1}$, **where** $M = \max|f^{(n+1)}(t)|$ ✓

b. $|R_n(x)| \leq \frac{M}{n!}|x-c|^n$

c. $|R_n(x)| \leq |f^{(n+1)}(c)|$

d. $|R_n(x)| \leq \frac{1}{(n+1)!}$

Taylor's Remainder Theorem states $|R_n(x)| \leq \frac{M}{(n+1)!}|x-c|^{n+1}$ where M is the maximum value of $|f^{(n+1)}(t)|$ on the interval between c and x.

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